

```
#include <stdio.h>

int main()
{
    int disk[100], n, fileSize, method;
    int i, j, start, count, indexBlock;

    printf("Enter total number of disk blocks: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++)
        disk[i] = 0;    // 0 = free, 1 = allocated

    printf("Enter file size (number of blocks): ");
    scanf("%d", &fileSize);

    printf("\n1. Contiguous\n");
    printf("2. Linked\n");
    printf("3. Indexed\n");
    printf("Enter allocation method: ");
    scanf("%d", &method);

    if (fileSize > n)
    {
        printf("Error: File size is larger than disk size.\n");
        return 0;
    }

    /* CONTIGUOUS */
    if (method == 1)
    {
        count = 0;
        start = -1;

        for (i = 0; i < n; i++)
        {
            if (disk[i] == 0)
                count++;
            else
                count = 0;

            if (count == fileSize)
            {
                start = i - fileSize + 1;
                break;
            }
        }
    }
}
```

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    if (start == -1)
    {
        printf("Error: Insufficient contiguous space.\n");
    }
    else
    {
        printf("Allocated blocks: ");

        for (i = start; i < start + fileSize; i++)
        {
            disk[i] = 1;
            printf("%d ", i);
        }

        printf("\n");
    }
}

/* LINKED */
else if (method == 2)
{
    count = 0;

    printf("Allocated blocks: ");

    for (i = 0; i < n && count < fileSize; i++)
    {
        if (disk[i] == 0)
        {
            disk[i] = 1;
            printf("%d ", i);
            count++;
        }
    }

    if (count < fileSize)
        printf("\nError: Not enough free blocks.\n");
    else
        printf("\n");
}

/* INDEXED */
else if (method == 3)
{
    if (fileSize + 1 > n)
    {
        printf("Error: Not enough blocks for index block and data.\n");
        return 0;
    }
}
```

```
/* Find index block */
indexBlock = -1;

for (i = 0; i < n; i++)
{
    if (disk[i] == 0)
    {
        indexBlock = i;
        disk[i] = 1;
        break;
    }
}

count = 0;

printf("Index block: %d\n", indexBlock);
printf("Allocated data blocks: ");

for (i = 0; i < n && count < fileSize; i++)
{
    if (disk[i] == 0)
    {
        disk[i] = 1;
        printf("%d ", i);
        count++;
    }
}

if (count < fileSize)
    printf("\nError: Not enough free blocks.\n");
else
    printf("\n");
}

else
{
    printf("Invalid allocation method.\n");
}

return 0;
}
```